Mixed strategies

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Game Theory and Mechanism Design - Lecture 5

(This lecture heavily relies on Tadelis (2013) Game Theory – An Introduction – chap. 6, and Narahari (2014) Game Theory and Mechanism Design – chap. 7)

Learning objectives

At the end of this lecture (and related tutorial(s)), a student should be able to:

- **Generalise** the concept of **strategies**
- **Recognise** and **justify** mixed-strategy Nash equilibria
- **Formulate** and **discuss** Nash existence theorem

Outline

1. Defining mixed strategies
2. Mixed strategy Nash equilibrium

1. **Defining mixed strategies**

From pure to mixed strategies

Pure strategies are deterministic (pure strategies were originally introduced in Lecture 2)

Definition (Pure strategy)

A pure strategy for player i is a deterministic plan of action (i.e., for a given information set as input, player i will always use the same action).
A profile of pure strategies $s = (s_1, \ldots, s_n)$ describes a combination of individual pure strategies for the n players in the games, $s_i \in S_i$, $\forall i$.

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Mixed strategies are probabilistic!

Definition (Mixed strategy)

For a player i with a set S_i of pure strategies, a *mixed strategy* σ_i is a probability distribution over S_i. It means that σ_i assigns a probability to play each of the pure strategies $s_i \in S_i$, i.e.

$$\sigma_i \,:\, S_i \to [0,1], \quad s.t. \quad \sum_{s_i \in S_i} \sigma_i(s_i) = 1$$

We write ΔS_i (the *simplex* of S_i) the set of all potential probability distributions over S_i, $\sigma_i \in \Delta S_i$.

Mixed strategies are also referred to as *randomised* strategies since, in practice, one actually randomly chooses the strategy to play (given the chosen probability distribution).

Illustration for matching pennies (without observation)

For matching pennies, we have the following matrix representation of the normal form game:

player 2

player 1		H	T
	H	$1, -1$	$-1, 1$
	T	$-1, 1$	$1, -1$

Pure strategies

$S_i = \{H,\ T\},\ i = 1, 2$

For player i, $i = 1, 2$:

- $\{\emptyset\} \to H$
- $\{\emptyset\} \to T$

Mixed strategies

$\Delta S_i = \{\sigma_i(H),\ \sigma_i(T)\},\ i = 1, 2$
such that $\sigma_i(H) + \sigma_i(T) = 1$

For player i, $i = 1, 2$:

- $\{\emptyset\} \to P[H] = \sigma_i(H),\ P[T] = \sigma_i(T)$

Rock-Paper-Scissors

You most likely know this game from your childhood years...

Here is the representation of the normal form game:

		player 2		
		R	P	S
player 1	R	$0, 0$	$-1, 1$	$1, -1$
	P	$1, -1$	$0, 0$	$-1, 1$
	S	$-1, 1$	$1, -1$	$0, 0$

R: Rock, **P**: Paper, **S**: Scissors

What would be **pure** and **mixed** strategies for that game?

Question...

Do these 2 games have pure-strategy Nash equilibria?

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player 2

player 1	H	T
H	$\underline{1, -1}$	$\overline{-1, 1}$
T	$\overline{-1, 1}$	$\underline{1, -1}$

No!

Question...

Do these 2 games have pure-strategy Nash equilibria?

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player 1	H	T
H	$\underline{1,-1}$	$\overline{-1,1}$
T	$\overline{-1,1}$	$\underline{1,-1}$

No!

player 2

player 1	R	P	S
R	$0,0$	$\overline{-1,1}$	$\underline{1,-1}$
P	$\underline{1,-1}$	$0,0$	$\overline{-1,1}$
S	$\overline{-1,1}$	$\underline{1,-1}$	$0,0$

No!

However, don't we intuitively know there is a best approach for both players, for which there would be a **concentration of best responses**?

Beliefs

Now that players can mix pure strategies (in a probabilistic sense), we need to generalise our definition of belief.

Definition (Belief)

For a player i in a normal form game, a *belief* is defined as a probability distribution $\pi_i \in \Delta S_{-i}$ over the strategies of their opponents. This is the way player i believes the opponents will play the game. Let us write $\pi_i(s_{-i})$ the probability player i puts on the specific strategy $s_{-i} \in S_{-i}$.

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	S	$-1,1$	$1,-1$	$0,0$

For instance of the RPS game:

- player 1 may believe that player 2 will play (R,P,S), with probability (0.2,0.5,0.3)
- player 2 may believe that player 1 will play (R,P,S), with probability (0.5,0.1,0.4)

Expected pay-off (1)

If we have a distribution over what others may play, what happen to the pay-offs?

Definition (Expected pay-off – pure vs. mixed)

For a player i choosing a pure strategy s_i while the opponents are using a mixed strategy σ_{-i}, the *expected pay-off* is the combination of pay-offs from application of the opponents' pure strategies, weighted by their corresponding probability, i.e.

$$u_i(s_i, \sigma_{-i}) = \sum_{s_{-i} \in \Delta S_{-i}} \sigma_{-i}(s_{-i})\, u_i(s_i, s_{-i})$$

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Example: if player 1 plays a pure strategy R, and player 2 plays a mixed strategy with probability (0.2,0.5,0.3),

$$\begin{aligned} u_1(R, \sigma_2) &= 0 \times 0.2 + (-1) \times 0.5 + 1 \times 0.3 \\ &= -0.2 \end{aligned}$$

Expected pay-off (2)

If we have a distribution over what others may play, what happen to the pay-offs?

Definition (Expected pay-off – mixed vs. mixed)

For a player i choosing a mixed strategy σ_i while the opponents are using a mixed strategy σ_{-i}, the *expected pay-off* is the combination of pay-offs over all pure strategies of both players, weighted by their corresponding probability, i.e.

$$u_i(\sigma_i, \sigma_{-i}) = \sum_{s_i \in \Delta S_i} \left(\sum_{s_{-i} \in \Delta S_{-i}} \sigma_i(s_i)\, \sigma_{-i}(s_{-i})\, u_i(s_i, s_{-i}) \right)$$

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Example: if player 1 plays a mixed strategy (0.5,0.1,0.4), and player 2 plays a mixed strategy with probability (0.2,0.5,0.3),

$$\begin{aligned} u_1(\sigma_1, \sigma_2) &= 0.5 \times (0 \times 0.2 + (-1) \times 0.5 + 1 \times 0.3) \\ &\quad + 0.1(1 \times 0.2 + 0 \times 0.5 + (-1) \times 0.3) \\ &\quad + 0.4(-1 \times 0.2 + 1 \times 0.5 + 0 \times 0.3) \\ &= -0.01 \end{aligned}$$

Support

When a player uses a mixed strategies, that does not mean they will mix all pure strategies...

Definition (Support of a mixed strategy)

For a player i with a set of pure strategies S_i, the *support* $\delta(\sigma_i)$ of a mixed strategy $\sigma_i \in \Delta S_i$ is the set of pure strategies used in the mixed strategies (that is, with non-zero probability), i.e.

$$\delta(\sigma_i) = \{s_i \in S_i : \sigma_i(s_i) > 0\}$$

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Examples:

- player 1 plays a mixed strategy (0.5,0,0.5), then the support is $\delta(\sigma_1) =$ (R,S)
- player 2 plays a mixed strategy (0,0.4,0.6), then the support is $\delta(\sigma_2) =$ (P,S)

Question

Based on what you have heard, do you think that **a pure strategy can also be seen as a mixed strategy?**

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Based on what you have heard, do you think that **a pure strategy can also be seen as a mixed strategy?**

The answer is **Yes!**

It is special case, which we refer to as a *degenerate mixed strategies*, since the support is reduced is a single pure strategy, played with a probability of 1...

2. **Mixed-strategy Nash equilibrium**

Best response

We can generalise the concept of best response to mixed strategies

Definition (Best response)

For a player i with potential mixed strategies $\sigma_i \in \Delta S_i$, given a given mixed strategy profile $\sigma_{-i} \in \Delta S_{-i}$ of the opponents, the *best response* strategy $b_i(\sigma_{-i})$ is defined as that which yields the highest expected pay-off, i.e.

$$b_i(\sigma_{-i}) = \{\sigma_i \in \Delta S_i \,:\, u_i(\sigma_i, \sigma_{-i}) \geq u_i(\sigma_i', \sigma_{-i}),\ \forall \sigma_i' \in \Delta S_i\}$$

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An additional important result builds on something we learned previously (i.e., a strictly dominated strategy can never be a best response)...

Proposition (Strictly dominated strategies, best response and support of mixed strategy)

For a player i, if one or more of their strategies $s_i \in S_i$ is strictly dominated, these strategies will never be part of the support of their best response strategy $b_i(\sigma_{-i})$, whatever the mixed strategy profile $\sigma_{-i} \in \Delta S_{-i}$ of the opponents

And now, remember the basic idea of the Nash equilibrium to look for a **concentration of best response**

Mixed-strategy Nash equilibrium

Then, we can similarly generalise the definition of Nash equilibria

Definition (Mixed-strategy Nash equilibrium - MSNE)

The mixed-strategy profile $\sigma^* = (\sigma_1^*, \ldots, \sigma_n^*)$ is a *mixed-strategy Nash equilibrium* if for each player i, σ_1^* is a best response to σ_{-i}^*. I.e., for all $i \in N$,

$$u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(\sigma_i, \sigma_{-i}^*), \quad \forall \sigma_i \in \Delta S_i$$

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Can you intuitively find the mixed-strategy Nash equilibria for the 2 games we looked at?

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Can you intuitively find the mixed-strategy Nash equilibria for the 2 games we looked at?

		player 2	
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player 1	H	$1, -1$	$-1, 1$
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Matching pennies: both players use the mixed strategy $(\frac{1}{2}, \frac{1}{2})$

		player 2		
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RPS: both players use the mixed strategy $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$

Another way to think about it

The above definition does not make it easy to identify such equilibria... But, we have something better!

Theorem (Necessary and sufficient conditions for MSNE)

The mixed-strategy profile $\sigma^* = (\sigma_1^*, \ldots, \sigma_n^*)$ is a mixed-strategy Nash equilibrium if and only if

(i) $u_i(s_i, \sigma_{-i}^*), \quad \forall s_i \in \delta(\sigma_i^*)$, *and*

(ii) $u_i(s_i, \sigma_{-i}^*) \geq u_i(s_i', \sigma_{-i}^*), \quad \forall s_i \in \delta(\sigma_i^*), \ \forall s_i' \notin \delta(\sigma_i^*)$

(Proof in Narahari (2014), pp. 99-100)

This is how to interpret it:

- For a given mixed strategy Nash equilibrium $\sigma^* = (\sigma_1^*, \ldots, \sigma_n^*)$, each player **gets the same pay-off** by playing any pure strategy in their support.
- Each player is then **indifferent** about which of the pure strategies (in their support) to play will play.
- To assess whether a mixed strategy profile is a Nash equilibrium, one can go through all pure strategies of a given player, given that the other players play their equilibrium strategies.

Illustration based on Rock-Paper-Scissors

For both players, they should use the mixed strategy $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$.

		player 2		
		R	P	S
	R	$0, 0$	$-1, 1$	$1, -1$
player 1	P	$1, -1$	$0, 0$	$-1, 1$
	S	$-1, 1$	$1, -1$	$0, 0$

Expected pay-offs:

- if player 1 plays R

$$\begin{aligned} u_1(R, \sigma_2) &= 0 \times \frac{1}{3} + (-1) \times \frac{1}{3} + 1 \times \frac{1}{3} \\ &= 0 \end{aligned}$$

- if player 1 plays P

$$\begin{aligned} u_1(P, \sigma_2) &= 1 \times \frac{1}{3} + 0 \times \frac{1}{3} + (-1) \times \frac{1}{3} \\ &= 0 \end{aligned}$$

- if player 1 plays S

$$\begin{aligned} u_1(S, \sigma_2) &= (-1) \times \frac{1}{3} + 1 \times \frac{1}{3} + 0 \times \frac{1}{3} \\ &= 0 \end{aligned}$$

Nash existence theorem

This was the core of his 28-page PhD thesis, which got him the Nobel Prize in Economics (1994)

Theorem

For any normal form game $\Gamma = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$, with finite number of players and finite strategy sets S_i for all players $i = 1, \ldots, n$, there exists at least one mixed-strategy Nash equilibrium.

(Proof in Nash's PhD thesis)

Some comments:

- Firstly, this is an **amazing** result!
- There is **always** one
- There is **at least** one, and then maybe more
- Since pure strategies are a special type of mixed strategies, it means that **these equilibria may include both pure and mixed-strategy equilibria**

Wrapping up

- **Strategies** and **equilibria** can now be appraised in both deterministic and probabilistic settings
- Nash existence theorem tells us that **there always exists at least 1 equilibrium!**
- The concepts of **dominance** and **IESDS** can be generalised to mixed-strategy equilibria (see Tadelis (2013) - section 6.3)
- In the general case (e.g., higher dimensions and more complex games), it can become **very difficult** to **find mixed-strategy equilibria**

Further readings (/lecture notes for this topic):

Tadelis S (2013) *Game Theory - An Introduction* - Chap. 6 (available on Blackboard Learn)
Narahari Y (2014) *Game Theory and Mechanism Design* - Chap. 7 (available on Blackboard Learn)

Thanks for your attention!